

WE – Gyaan Interactive Science Lab

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A comprehensive mapping of educational exhibits aligned with NCERT curriculum concepts across Grades 5–10 of the Interactive Science Labs, a project of WE Samrt Labs.



Science Exhibits Overview

Sr. No.	Exhibit Name	Picture of Exhibit	Concepts Covered	5th	6th	7th	8th	9th	10th	Electric/Non-Electric
1	Day and Night Model		Day and Night, Shadow Formation		11. Sun in Space				17. Stars and the Solar System	Electric
2	Sundial		Measurement of Time			13. Motion and Time				Non-Electric
3	Student Time Indicator		Measurement of Time at Different Locations			13. Motion and Time				Electric
4	Light Travels in a Straight Line		Path of Light		11. Light, Shadows and Reflection				15. Light	Non-Electric
5	Colour Shadows		Formation of Secondary Colours from Primary Colours, Formation of White Light		11. Light, Shadows and Reflection					Electric
6	Newton's Disc		Combination of Colours						15. Light	Electric
7	Virtual Time Sensor						13. Sound			Electric
8	Tuning Fork (8 Forks)		Sound Formation, Vibration, High and Low Pitch				13. Sound		12. Sound	

Reflection and Optical Exhibits

S. No.	Exhibit Name	Exhibit Picture	Key Concepts	5th	6th	7th	8th	9th	10th	Electric/N on-Electric
7	Law of Reflection		Reflection, Laws of Reflection				16. Light		10. Light - Reflection and Refraction	
8	Kaleidoscope		Multiple Reflections, Different Patterns		11. Light, Shadows, and Reflection		16. Light			
9	Periscope		Path of Light, Reflection of Light		11. Light, Shadows, and Reflection		16. Light			
10	Corner Mirror Magic		Image Formation with Respect to Angle				16. Light			

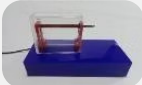
Mirrors, Lenses and Advanced Optics

Sr. No.	Exhibit Name	Picture of Exhibit	Concepts Illustrated by the Exhibit	5th	6th	7th	8th	9th	10th	Electric/Non-Electric
11	Fun Mirrors		Reflection, Concave and Convex Mirrors						10. Light - Reflection and Refraction	
12	Concave and Convex Lenses		Image formation with concave and convex lenses				15. Light		10. Light - Reflection and Refraction	
13	Total Internal Reflection		Total Internal Reflection, Critical Angle						10. Light - Reflection and Refraction	
14	Infinity Well		Multiple Reflection, Infinite Images			15. Light	16. Light		10. Light - Reflection and Refraction	Electric

Vision and Magnetism Exhibits

Sr. No.	Exhibit Name	Picture of Exhibit	Concepts Illustrated by the Exhibit	5th	6th	7th	8th	9th	10th	Electric/Non-Electric
15	Angular Vision		Field of Vision in Animals						11. The Human Eye and the Colorful World	Electric
16	Persistence of Vision		Persistence of Vision				16. Light		11. The Human Eye and the Colorful World	Electric
17	Depth Perception		2D and 3D Images in Vision				16. Light		11. The Human Eye and the Colorful World	Non-Electric
18	Magnetic Field Visualization		Polar Properties of Magnets, Lines of Magnetic Field		13. Fun with Magnets					Non-Electric
19	Magnetic Spring		Polar Properties of Magnets, Attraction and Repulsion		13. Fun with Magnets					Non-Electric

Electromagnetism Exhibits

Sr. No.	Exhibit Name	Picture of Exhibit	Concepts Aligned with the Exhibit	5th	6th	7th	8th	9th	10th	Electric/Non-electric
20	Electromagnet		Effect of electric current on metal, Electromagnetism				14. Electric Current and Its Effects			Electric
21	Magnetic Crane		Effect of electric current on metal, Electromagnetism		13. Fun with Magnets					Electric
22	Electromagnetic Bell		Electromagnetism				14. Electric Current and Its Effects			Electric
23	3D Magnetic Lines		Polar properties of magnets, Lines of magnetic field		13. Fun with Magnets			13. Magnetic Effect of Electric Current		Non-electric

Electricity and Circuits

Sr. No.	Exhibit Name	Picture of Exhibit	Concepts Aligned with Exhibit	5th	6th	7th	8th	9th	10th	Electric/N on-electric
24	Series Circuit		Types and Arrangement of Circuits						12. Electricity	electric
25	Parallel Circuit		Types and Arrangement of Circuits						12. Electricity	electric
26	Good and Bad Conductors		Conductivity in Matter		12. Electricity and Circuits					electric
27	Simple Circuit		Types and Arrangement of Circuits		12. Electricity and Circuits				12. Electricity	electric
28	Simple Motor		Circuits, Electromagnetism					13. Magnetic Effect of Electric current		electric
29	Two-Way Switch		Types of Switch		12. Electricity and Circuits					electric
30	Ohm's Law		Relation between Potential Difference and Current						12. Electricity	electric

Forces, Energy and Motion

Sr. No.	Exhibit Name	Picture of Exhibit	Concepts Aligned with the Exhibit	5th	6th	7th	8th	9th	10th	Electric/Non-Electric
31	Floating Ball		Bernoulli's Principle, High Pressure - Low Pressure				8. Winds, Storms, and Cyclones			Electric
32	Hand Pump		Drawing Water from Underground				16. Water: A Precious Resource			
33	Centrifuge Marbles		Centrifugal Force		11. Sun, Earth and Space					
34	Pisa Tower		Centre of Gravity						10. Gravitation	
35	Windmill		Source of Energy, Conversion of Wind to Electric Energy	15. Air Around Us					14. Sources of Energy	Electric
36	Solar Fan		Source of Energy, Conversion of Solar to Electric Energy						14. Sources of Energy	Electric
37	Tornado		Formation of Cyclones in Nature				8. Wind, Storms, and Cyclones			Non-Electric

Mechanics and Pendulum Exhibits

Sr. No.	Exhibit Name	Picture of Exhibit	Concepts Related to the Exhibit	5th	6th	7th	8th	9th	10th	Electric/Non-electric
38	Newton's Cradle		Conservation of Momentum					9. Force and Laws of Motion		
39	Simple Pendulum		Oscillation, Time Period			13. Motion and Time	13. Sound			
40	Coupled Pendulum		Conversion and Conservation of Energy			13. Motion and Time				
41	Cycloid Path		Relationship of Potential and Kinetic Energy with Height				8. Motion			
42	Looping the Loop		Potential Energy at Height				8. Motion	11. Work and Energy		
43	Gyroscope		Effect of Velocity on Direction				8. Motion	11. Work and Energy		

Newton's Laws and Balance Exhibits

Sr. No.	Exhibit Name	Picture of Exhibit	Concepts Aligned with the Exhibit	5th	6th	7th	8th	9th	10th	Electric/Non-Electric
47	Newton's First Law		First Law of Motion					9. Force and Laws of Motion		
48	Newton's Second Law		Newton's Second Law					9. Force and Laws of Motion		
49	Newton's Third Law		Action-Reaction					9. Force and Laws of Motion		
50	Balancing Doll		Centre of Gravity						10. Gravitation	
51	Balancing Nails		Centre of Gravity						10. Gravitation	

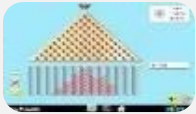
Simple Machines and Motion Exhibits

Sr. No.	Exhibit Name	Picture of Exhibit	Concepts Aligned with the Exhibit	5th	6th	7th	8th	9th	10th	Electric/Non-Electric
52	Pulleys		Simple Machines					11. Work and Energy		Non-Electric
53	Levers		Simple Machines					11. Work and Energy		Non-Electric
54	Wheel & Axle		Simple Machines					11. Work and Energy		Non-Electric
55	Archimedes' Screw		Simple Machines					11. Work and Energy		Non-Electric
56	Circular to Oscillatory Motion		Types of Motion				8. Motion			Non-Electric
57	Density Cubes		Relative Density of Material			16. Experiments with Water			10. Gravitation	Non-Electric

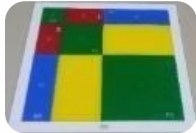
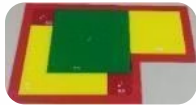
Biology and Human Body Models

Sr. No.	Exhibit Name	Picture of Exhibit	Concepts Aligned with the Exhibit	5th	6th	7th	8th	9th	10th	Electric/Non-Electric
58	Human Skeleton 85 cm XC-102B		Bones, Joints				8. Body Movements			
59	Heart Model ZX1405E		Blood Circulation, Heart Function						6. Life Processes	
60	DNA		Structure of DNA					9. Heredity and Evolution		
61	Kidney Model ZX-1407		Excretion						6. Life Processes	
62	Brain Parts XC-308		Nervous System, Parts of the Brain					7. Control and Coordination		
63	Ear Model XC-303C		Parts of the Ear				13. Sound		12. Sound	
64	Human Torso 23/11 50 cm MK-050		Organs in Body, Digestion, Respiration, Circulation			2. Nutrition in Animals, 10. Respiration in Organisms, 11. Transportation in Animals and Plants			6. Life Processes	

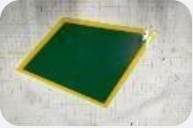
Basic Mathematics Models and Puzzles

Sr. No.	Exhibit Name	Picture of Exhibit	Concepts Aligned with the Exhibit	5th	6th	7th	8th	9th	10th	Electric/Non-Electric
65	Pythagoras		Relation of two sides of a triangle with the hypotenuse						6. Triangles	
66	Tower of Hanoi		Problem-solving with algorithms	To all grades						
67	Probability		Probability					Probability	15. Probability	
68	Tangram		Puzzle with different shapes	4. Parts and Wholes						
69	Equilateral Triangle to Square Puzzle		Puzzle with different shapes			6. Lines and Angles				
70	Two Congruent Right Triangles		Area of different shapes			7. Congruence of Triangles		7. Triangles		
71	LCM		Lowest Common Multiple		3. Playing with Numbers				1. Real Numbers	
72	$(a+b)^2 = a^2 + 2ab + b^2$		Polynomials				9. Algebraic Expressions and Identities	Polynomials		

Algebraic Identities Models

Sr. No.	Exhibit Name	Picture of Exhibit	Concepts Aligned with the Exhibit	5th	6th	7th	8th	9th	10th	Electric/Non-Electric
73	$(A + B)^3 = A^3 + 3A^2B + 3AB^2 + B^3$		Polynomials				9. Algebraic Expression and Identities	Polynomials		
74	$(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$		Polynomials				9. Algebraic Expression and Identities	Polynomials		
75	$(a + b)^2 - (a - b)^2 = 4ab$		Polynomials				9. Algebraic Expression and Identities			
76	$(A + B)^2 + (A - B)^2 = 2A^2 + 2B^2$		Polynomials				9. Algebraic Expression and Identities			

Geometry and Shapes Models

Sr. No.	Exhibit Name	Picture of Exhibit	Concepts align with the exhibit	5th	6th	7th	8th	9th	10th	Electric/Non-electric
77	Sum of Angles of a Quadrilateral		Sum of Angles of a Quadrilateral = 360°				Understanding Quadrilaterals	Quadrilaterals		
78	Fibonacci Series		Every term in the sequence is the sum of terms before it.							
79	Area of a Circle		Area of different shapes						12. Area Related to Circles	
80	(A2 - B2)		Algebraic expression				9. Algebraic expression and Identities			